Fundamentals of AI and KR - Module 3

2. Bayesian network representation

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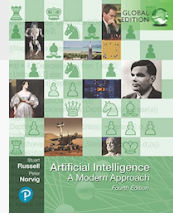
Fall 2026

Independence

Key idea

Independence identifies situations in which learning the value of one variable does not change an agent's beliefs about another variable.

Rather than specifying a probability for every possible complete assignment, independence allows us to factorize a joint distribution into smaller probability distributions. This can dramatically reduce the number of parameters needed to represent uncertain knowledge.



Marginal (or unconditional) independence I

Marginal independence

Two events A and B are independent, written $A \perp B$, if learning whether B occurs does not change the agent's belief in A:

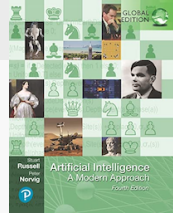
$$P(A \mid B) = P(A), \qquad \text{provided that } P(B) > 0,$$

and, equivalently,

$$P(B \mid A) = P(B), \qquad \text{provided that } P(A) > 0.$$

Equivalently:

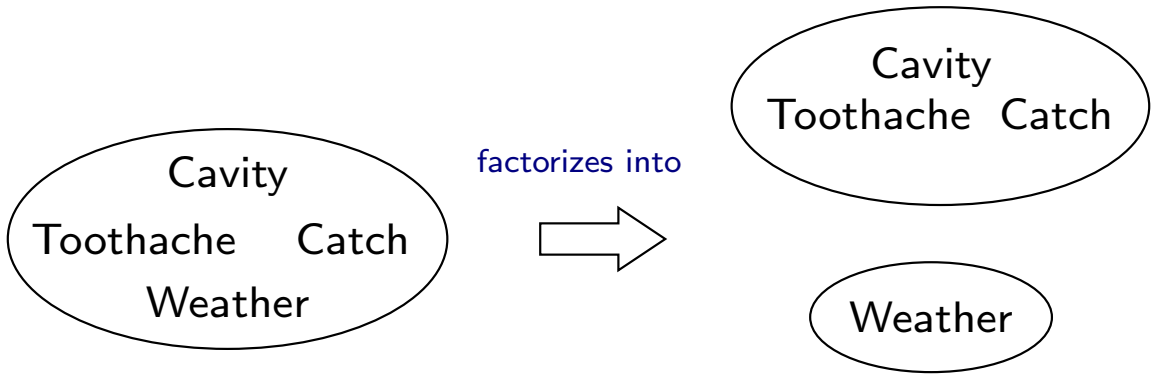
$$P(A \wedge B) = P(A)\,P(B).$$



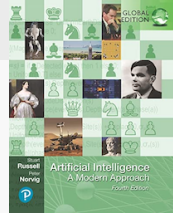
Marginal (or unconditional) independence II

Suppose that *Weather* is independent of the dental variables:

$$Weather \perp (Toothache, Catch, Cavity).$$



$$P(Toothache, Catch, Cavity, Weather) = P(Toothache, Catch, Cavity)\, P(Weather).$$



Marginal (or unconditional) independence III

Representation benefit

The full joint distribution has

$$2 \times 2 \times 2 \times 4 = 32$$

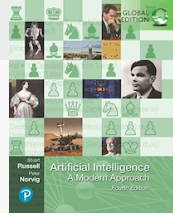
entries, corresponding to $32 - 1 = 31$ free parameters.

Under the independence assumption, it factorizes into:

$$\underbrace{(2^3 - 1)}_{\substack{P(Toothache, Catch, Cavity) \\ 7 \text{ free parameters}}} + \underbrace{(4 - 1)}_{\substack{P(Weather) \\ 3 \text{ free parameters}}} = 10$$

free parameters.

Equivalently, we store $2^3 + 4 = 12$ probability values rather than 32.



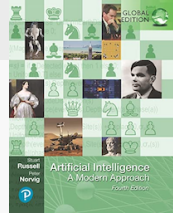
Marginal (or unconditional) independence IV

For n mutually independent Boolean variables, only n parameters are needed, rather than $2^n - 1$.

Marginal independence is powerful, but rare in realistic domains.

Dentistry is a large field with hundreds of variables, none of which are independent. What to do?

We next consider conditional independence.



Conditional independence I

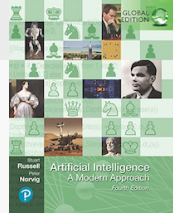
Conditional independence

Two variables X and Y are conditionally independent given Z, written

$$X \perp Y \mid Z,$$

if, once Z is known, learning the value of Y does not change the agent's probability distribution over X:

$$P(X \mid Y, Z) = P(X \mid Z).$$



Conditional independence II

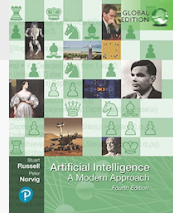
In the dental domain: $Catch \perp Toothache \mid Cavity$.
This means that, once we know whether the patient has a cavity, knowing whether they have a toothache provides no additional information about whether the probe catches:

$$P(Catch \mid Toothache, Cavity) = P(Catch \mid Cavity).$$

For example: $P(catch \mid toothache, cavity) = P(catch \mid cavity)$.
The same holds when $Cavity = false$.

Important distinction

Catch and *Toothache* may be dependent in general. They become independent only after conditioning on *Cavity*.



Conditional independence: Equivalent Formulation I

Symmetry of conditional independence

$$P(\textbf{Catch} \mid \textbf{Toothache}, \mathit{Cavity}) = P(\textbf{Catch} \mid \mathit{Cavity})$$

$$\Updownarrow$$

$$P(\textbf{Toothache} \mid \textbf{Catch}, \mathit{Cavity}) = P(\textbf{Toothache} \mid \mathit{Cavity})$$

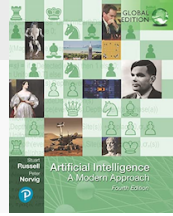
Factorization

It can also be written as a factorization:

$$P(\mathit{Toothache}, \mathit{Catch} \mid \mathit{Cavity}) = P(\mathit{Toothache} \mid \mathit{Cavity})\, P(\mathit{Catch} \mid \mathit{Cavity}).$$

Notation:

$$\mathit{Catch} \perp \mathit{Toothache} \mid \mathit{Cavity}.$$



Conditional independence and factorization I

Without independence assumptions, the joint distribution over 3 Boolean variables has: $8 - 1 = 7$ free parameters
For the chain rule:

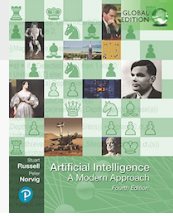
$$P(Cavity, Toothache, Catch) = P(Cavity)\, P(Toothache \mid Cavity) \cdot P(Catch \mid Toothache, Cavity).$$

Because $Catch \perp Toothache \mid Cavity$, we can simplify the last term:

$$\boxed{P(Cavity, Toothache, Catch) = P(Cavity)\, P(Toothache \mid Cavity) \cdot P(Catch \mid Cavity).}$$

What changed?

Once *Cavity* is known, *Toothache* provides no additional information about *Catch*.



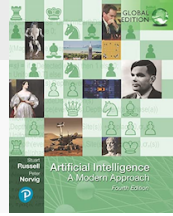
Conditional independence and factorization II

The factorized representation requires:

Distribution	**Free parameters**
$P(Cavity)$	1
$P(Toothache \mid Cavity)$	2
$P(Catch \mid Cavity)$	2
Total	**5**

7 free parameters $\longrightarrow$ 5 free parameters

Conditional independence lets us replace a joint distribution with smaller conditional distributions.

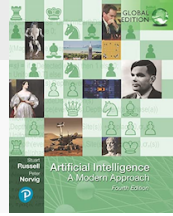


Conditional independence and factorization III

Why conditional independence matters

Conditional independence captures **local relevance**: once the directly relevant variables are known, many other variables provide no additional information.

In most cases, the use of conditional independence reduces the size of the representation of the joint distribution from exponential in n to linear in n.



Reversing conditional probability

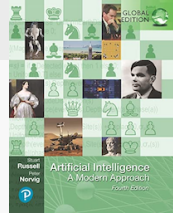
Medical tests are commonly described by conditional probabilities such as:

$$P(+ \mid \mathit{Disease}) \qquad \text{and} \qquad P(+ \mid \neg \mathit{Disease}).$$

For example, $P(+ \mid \mathit{Disease})$ is the probability that the test is positive **if the patient has the disease**. After observing a positive test result, however, the question of interest is:

$$\boxed{P(\mathit{Disease} \mid +)}$$

The probability of a positive result given a disease is not the same as the probability of a disease given a positive result.
Bayes' rule allows us to **reverse** the conditioning.



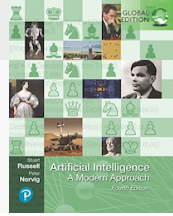
Bayes' rule I

Using the product rule, the joint event "the patient has the disease and the test is positive" can be factorized in two ways:

$$\begin{aligned} P(Disease, +) &= P(+ \mid Disease)\, P(Disease) \\ &= P(Disease \mid +)\, P(+). \end{aligned}$$

Solving gives Bayes' rule:

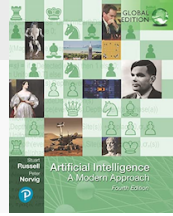
$$\boxed{P(Disease \mid +) = \frac{P(+ \mid Disease)\, P(Disease)}{P(+)}} \qquad P(+) > 0.$$



Bayes' rule II

How to read the formula

$$\underbrace{P(Disease \mid +)}_{\text{posterior}} = \frac{\underbrace{P(+ \mid Disease)}_{\text{test result if disease is present}} \cdot \underbrace{P(Disease)}_{\text{prior belief}}}{\underbrace{P(+)}_{\text{overall probability of a positive result}}}.$$



Bayes' rule: general form I

The medical-test example illustrates a general rule. For any two events A and B, with $P(B) > 0$:

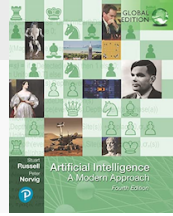
$$\boxed{P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}}$$

How to read the rule

Bayes' rule updates the prior belief $P(A)$ after observing B, using how compatible the observation is with A, namely $P(B \mid A)$.

For random variables Y and X, and an observed value $X = x$:

$$P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)\, P(Y = y)}{P(X = x)}.$$



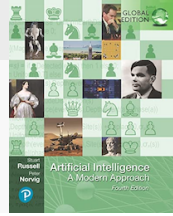
Bayes' rule: general form II

Equivalently, for the whole posterior distribution over Y:

$$P(Y \mid X = x) = \alpha\, P(X = x \mid Y)\, P(Y), \qquad \alpha = \frac{1}{P(X = x)}.$$

Normalization

The constant α ensures that the resulting probabilities over all possible values of Y sum to 1.

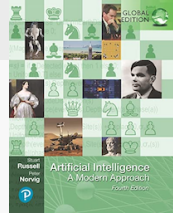


A diagnostic example: meningitis I

Let M denote the event "the patient has meningitis" and S the event "the patient has a stiff neck".

Suppose we know that:

- $P(M) = 1/50{,}000$: one person in every 50,000 has meningitis;
- $P(S) = 0.01$: 1% of individuals have a stiff neck;
- $P(S \mid M) = 0.70$: among individuals with M, 70% have a stiff neck.



A diagnostic example: meningitis II

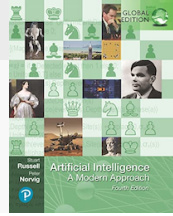
Question

A patient has a stiff neck. What is the probability that they have meningitis?

$$P(M \mid S).$$

Using Bayes' rule:

$$\begin{aligned} P(M \mid S) &= \frac{P(S \mid M)\, P(M)}{P(S)} \\ &= \frac{0.70 \cdot (1/50{,}000)}{0.01} \\ &= 0.0014 = 0.14\%. \end{aligned}$$



A diagnostic example: meningitis III

Note: the prior matters!!

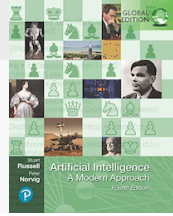
A stiff neck is common among patients with meningitis:

$$P(S \mid M) = 0.70.$$

However, meningitis is very rare in the overall population. Therefore,

$$P(S \mid M) = 0.70 \qquad \text{does not imply} \qquad P(M \mid S) = 0.70.$$

Observing the symptom increases the probability of meningitis from 0.002% to 0.14%, but the posterior probability remains low.



Bayes' rule with conditionally independent evidence I

Suppose that we observe both a toothache and a positive probe catch. We want to update the probability of a cavity:

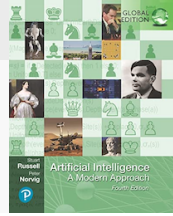
$$P(cavity \mid toothache, catch).$$

By Bayes' rule:

$$P(cavity \mid toothache, catch) = \alpha\, P(toothache, catch \mid cavity)\, P(cavity).$$

Conditional independence tells us that:

$$P(toothache, catch \mid cavity) = P(toothache \mid cavity)\, P(catch \mid cavity).$$

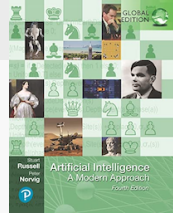


Bayes' rule with conditionally independent evidence II

Therefore:

$$P(cavity \mid toothache, catch) = \alpha\, P(toothache \mid cavity)\, P(catch \mid cavity) \cdot P(cavity)$$

Conditional independence lets us avoid specifying a probability for every combination of observations.

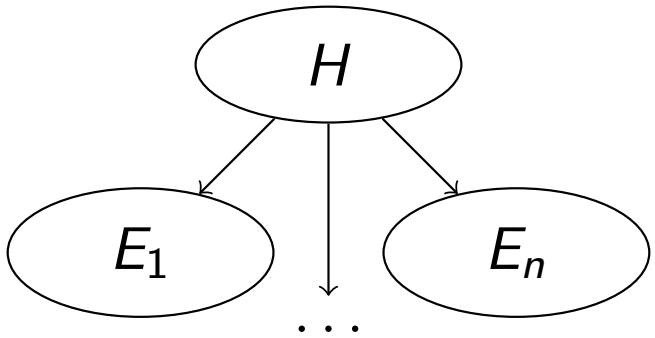


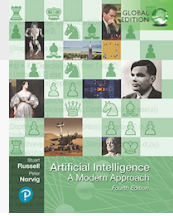
Naive Bayes I

Naive Bayes model

A naive Bayes model consists of one hypothesis variable H and a set of observation variables $E_1, \ldots, E_n$. It assumes that the observations are conditionally independent given H:

$$P(H, E_1, \ldots, E_n) = P(H) \prod_{i=1}^{n} P(E_i \mid H).$$





Naive Bayes II

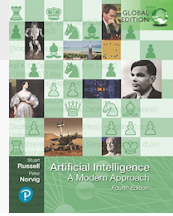
Example

In the dental example, $H = Cavity$, $E_1 = Toothache$, and $E_2 = Catch$.

If H has k values and each observation has at most r values, the model requires

$$(k-1) + nk(r-1)$$

free parameters: **linear in the number n of observations**, for fixed k and r.



Summary so far

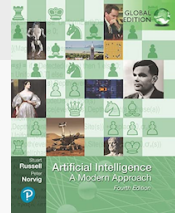
- Probability provides a rigorous formalism for representing and reasoning about uncertain knowledge.
- A joint probability distribution over a set of variables assigns a probability to every atomic event.
- In principle, any probability query can be answered by summing the atomic events compatible with the query and, when needed, normalizing the result.
- For nontrivial domains, a full joint distribution is too large to represent explicitly.
- Independence and conditional independence allow us to factorize the joint distribution into smaller probability distributions.
- Bayes' rule updates beliefs about a hypothesis after evidence is observed; naive Bayes is a simple model that exploits conditional independence to do so efficiently.

Bayesian network representation

Key idea

A Bayesian network represents a joint probability distribution compactly by combining a directed graph with local conditional probability distributions.

The graph makes conditional-independence assumptions explicit and determines how the full joint distribution factorizes.



Bayesian networks I

Bayesian network: syntax

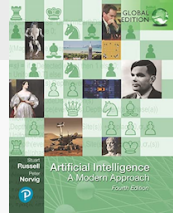
A Bayesian network for a set of random variables $X_1, \ldots, X_n$ consists of:

- one node for each variable;
- a directed acyclic graph (DAG), i.e., a graph with directed edges and no directed cycles;
- one local conditional probability distribution for each node:

$$P(X_i \mid \mathrm{Parents}(X_i)).$$

An edge $X_j \to X_i$ means that X_j is a parent of X_i: the local distribution of X_i is conditioned on X_j.

The edge represents a direct probabilistic dependency in the model; it does not imply a causal relationship (link $\approx$ "directly influences").
$\Rightarrow$ Topology of network encodes conditional independence assertions

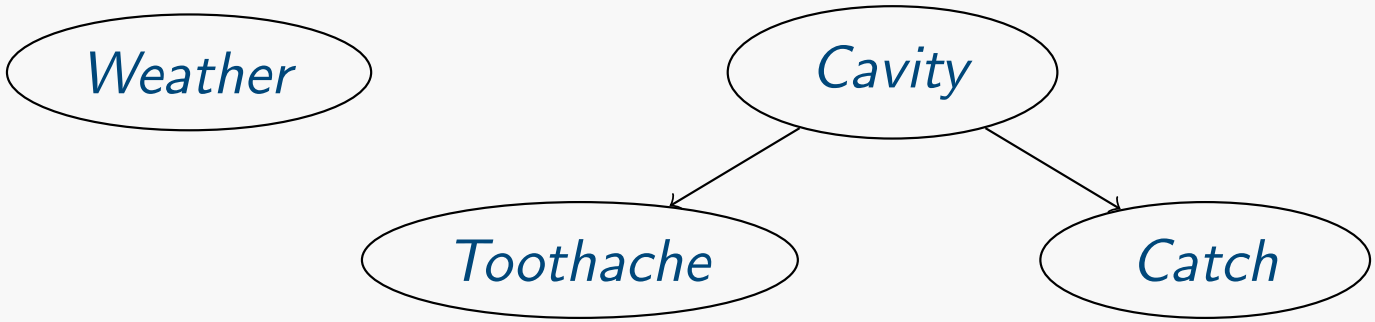


Bayesian networks II

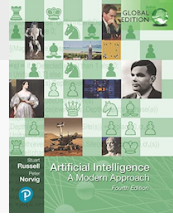
Conditional probability tables

When the variables are discrete, each local distribution can be represented by a conditional probability table (CPT). For every combination of parent values, the corresponding row gives a probability distribution over the possible values of X_i.

Dental use case



- *Weather* is independent of the other variables
- *Toothache* and *Catch* are conditionally independent given *Cavity*



Example I

You are at work when your neighbour John calls to say that your home alarm is ringing. Your other neighbour, Mary, does not call. The alarm may have been triggered by a burglary, but occasionally also by a minor earthquake.

Variables

$$Burglary, \quad Earthquake, \quad Alarm, \quad JohnCalls, \quad MaryCalls.$$

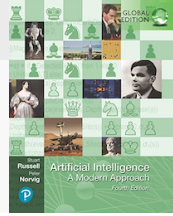
Evidence and query

The observed evidence is:

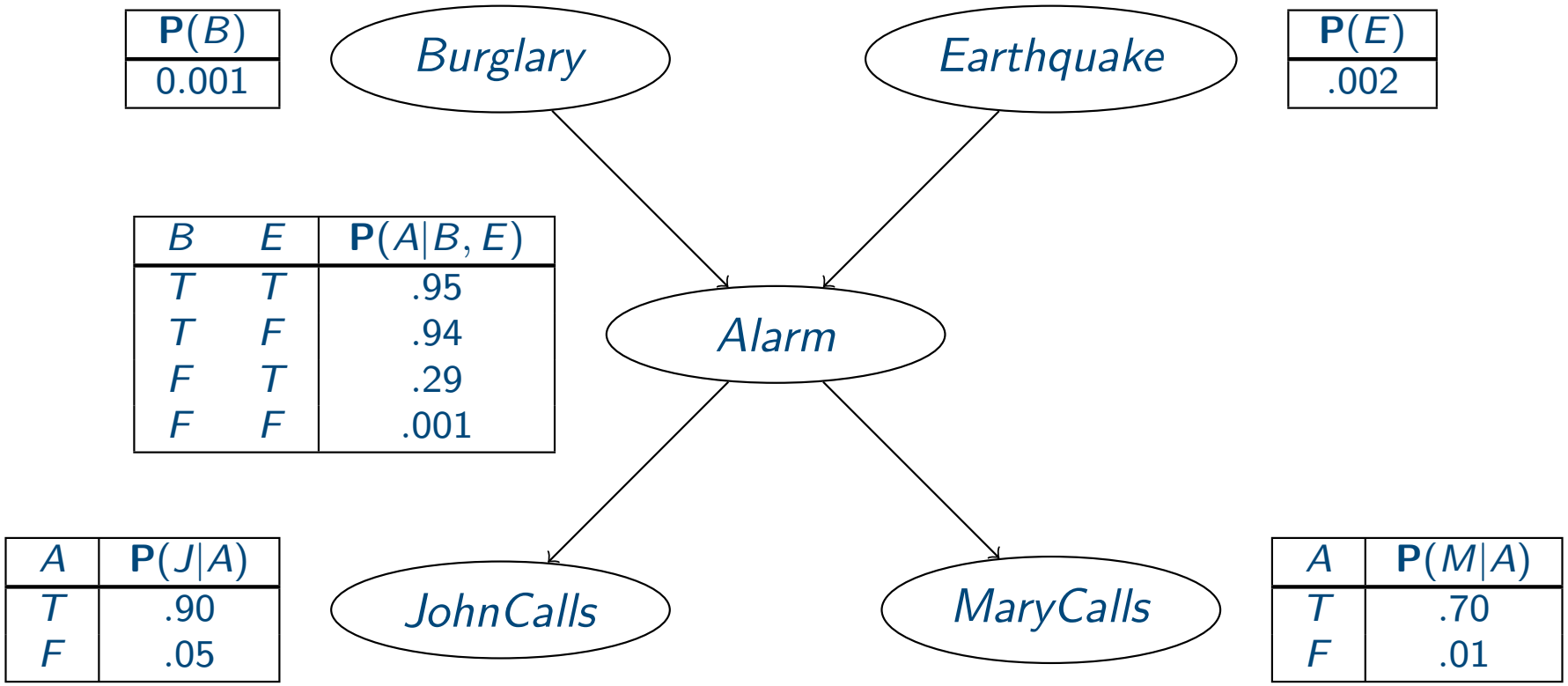
$$JohnCalls = true, \qquad MaryCalls = false.$$

We want to answer:

$$\boxed{P(Burglary = true \mid JohnCalls = true,\ MaryCalls = false)}$$



Example II



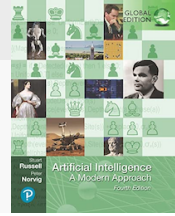
$\mathbf{P}(B)$
0.001

$\mathbf{P}(E)$
.002

B	E	$\mathbf{P}(A\|B,E)$
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	$\mathbf{P}(J\|A)$
T	.90
F	.05

A	$\mathbf{P}(M\|A)$
T	.70
F	.01



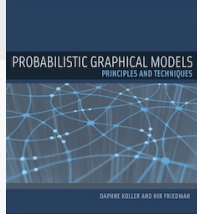
Reasoning with Bayesian networks

A Bayesian network compactly represents a joint probability distribution. Once its local probability distributions are specified, the network can answer probability queries of the form:

$$P(Query \mid Evidence).$$

Inference does not have to follow the direction of the arrows.

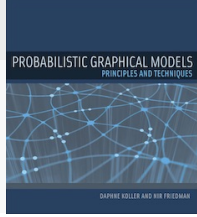
We can reason forward from possible causes to consequences, backward from observations to possible explanations, or between alternative explanations of the same observation.



Reasoning patterns

The student network

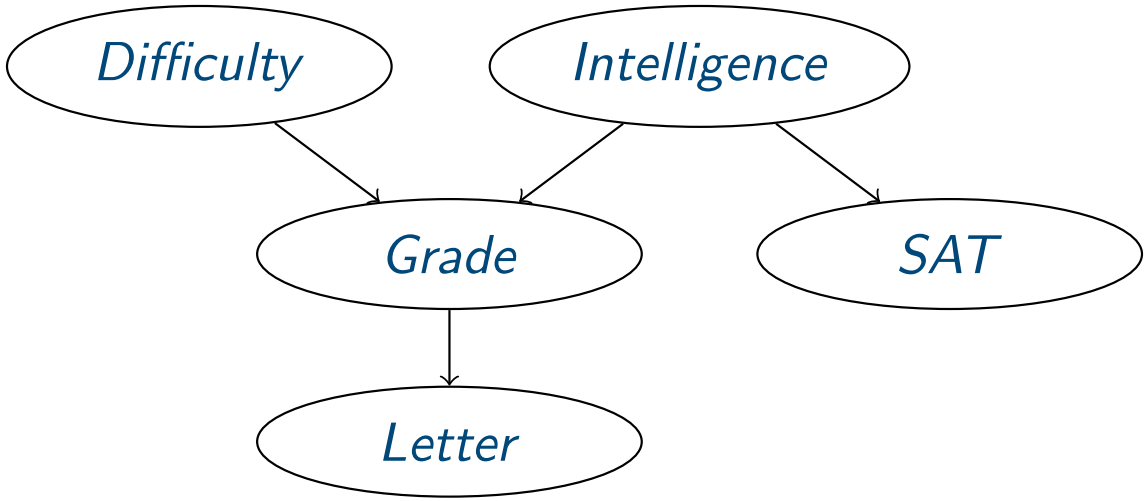
A student's grade depends on intelligence and on the difficulty of the course. SAT scores are correlated with intelligence. A professor writes recommendation letters by only looking at grades.

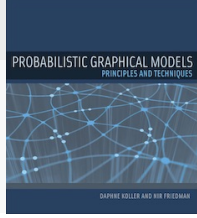


Reasoning patterns

The student network

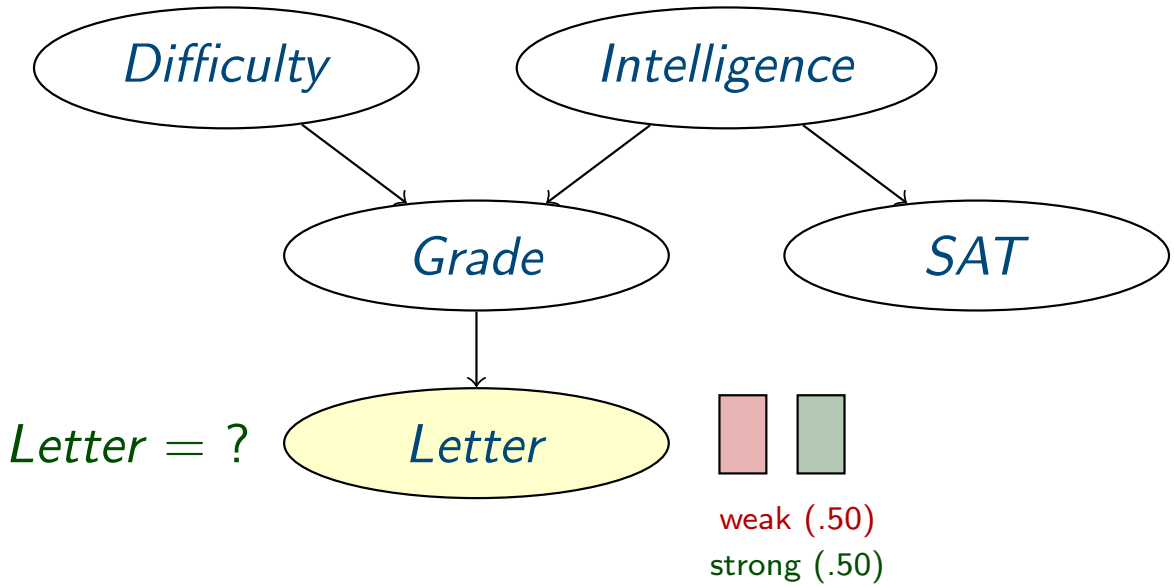
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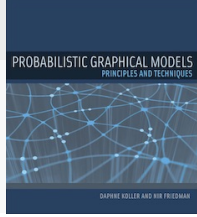




Reasoning patterns

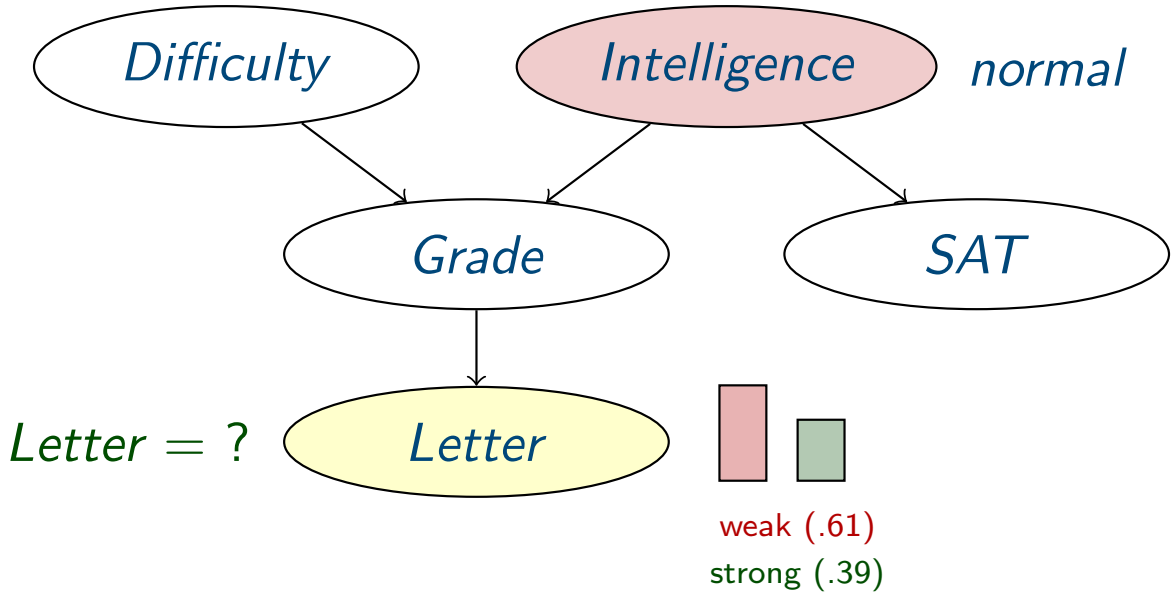
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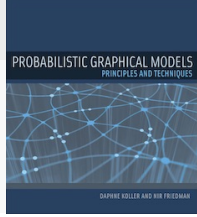




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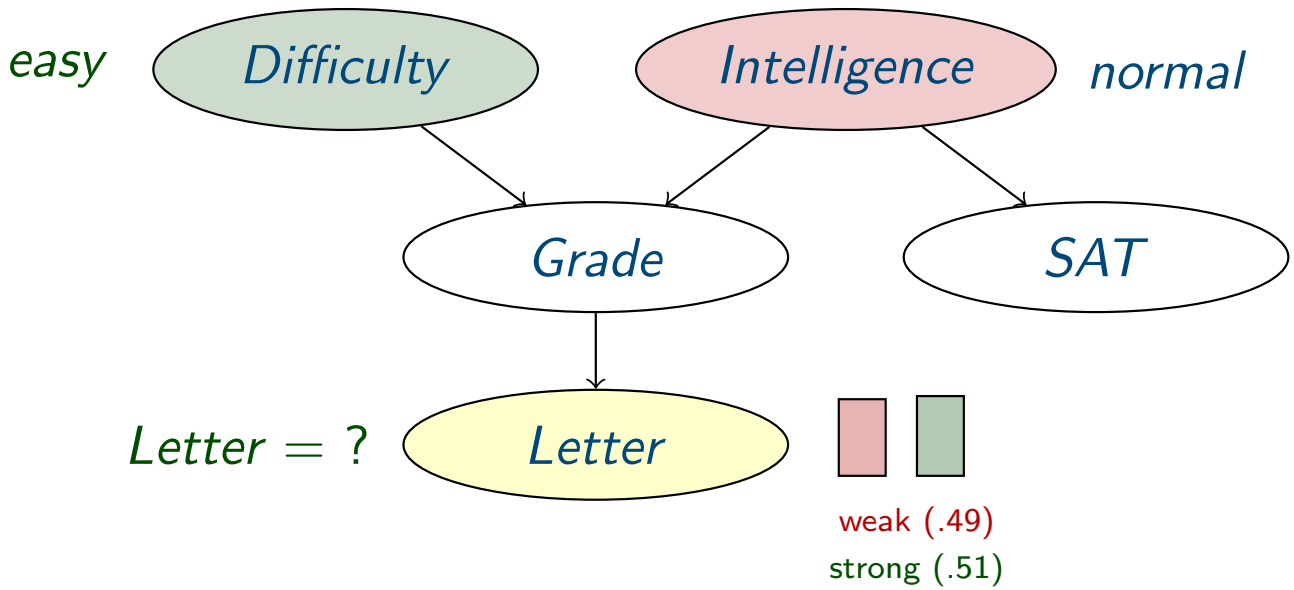
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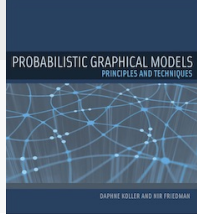




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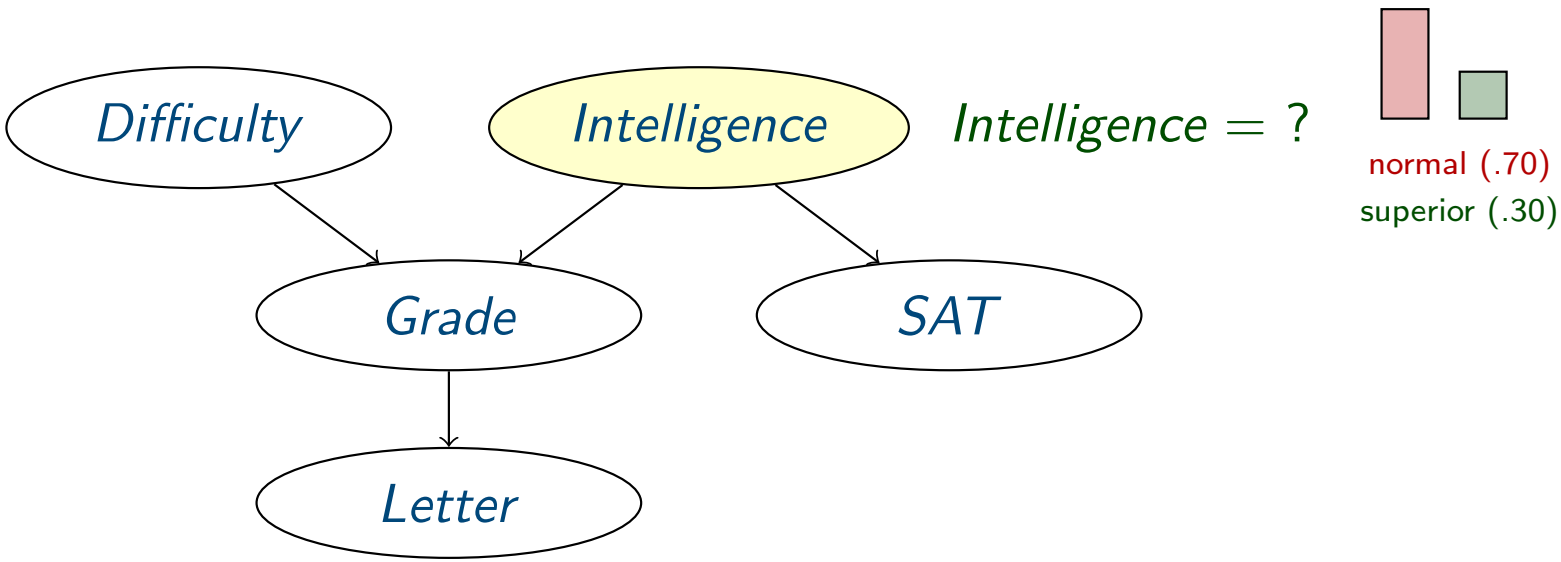
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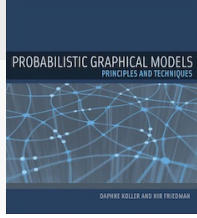




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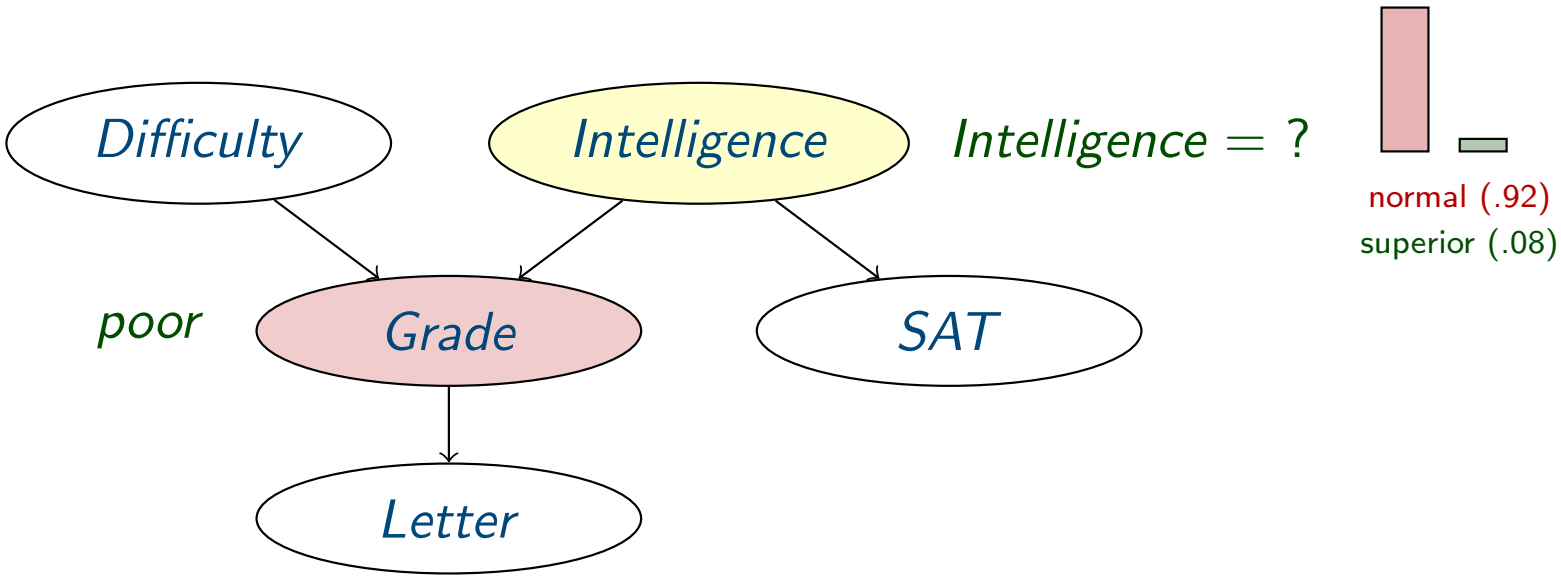
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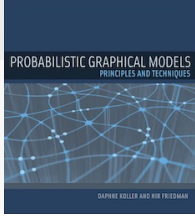




Reasoning patterns

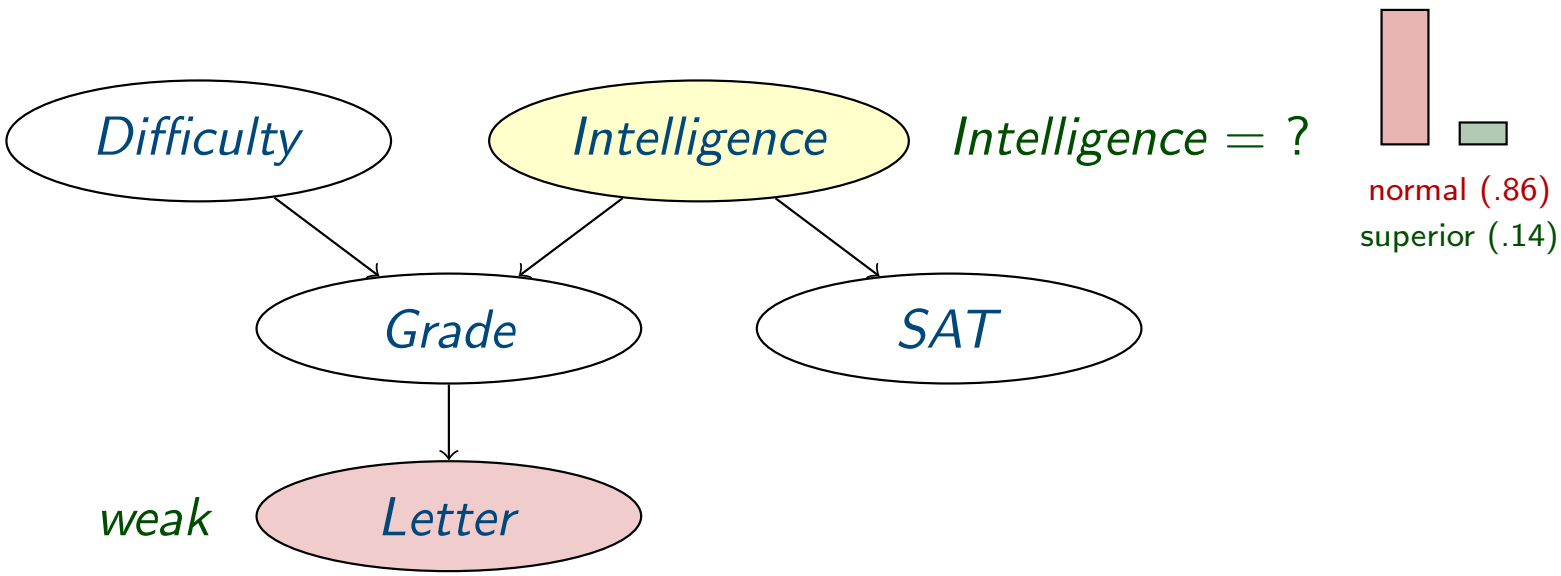
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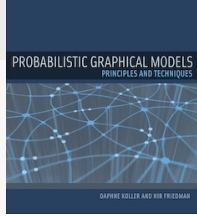




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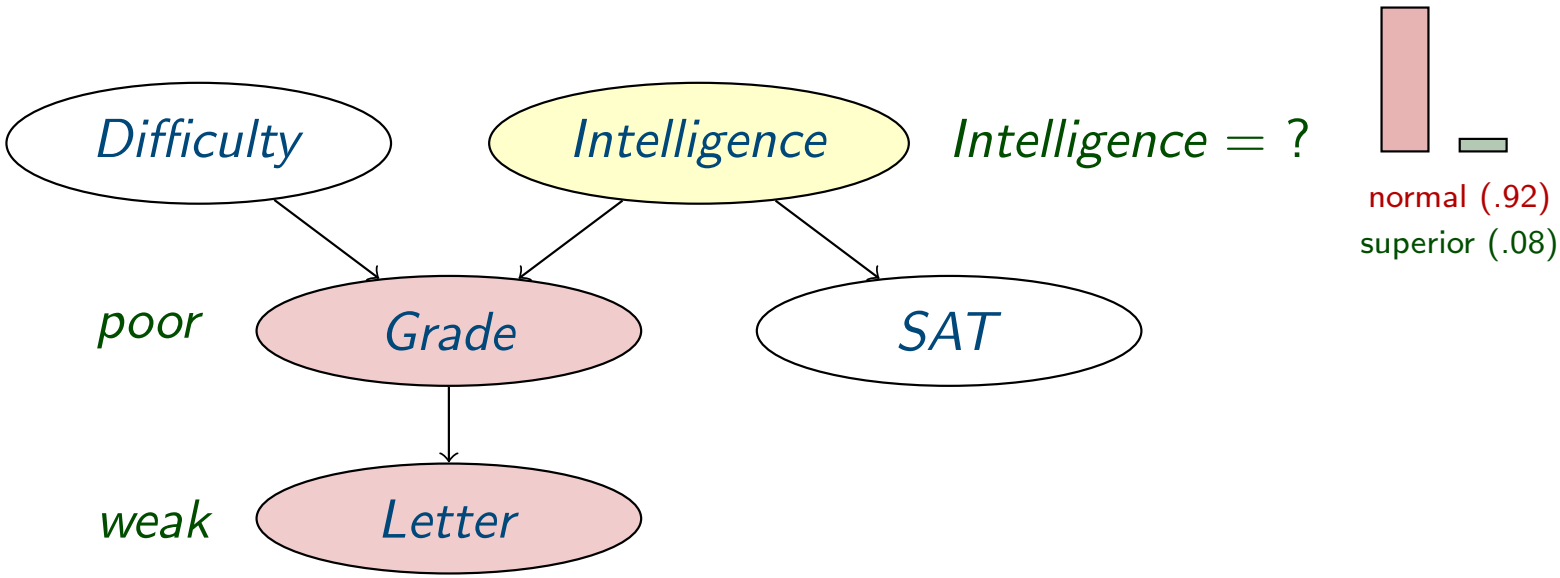
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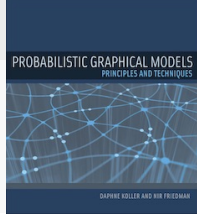




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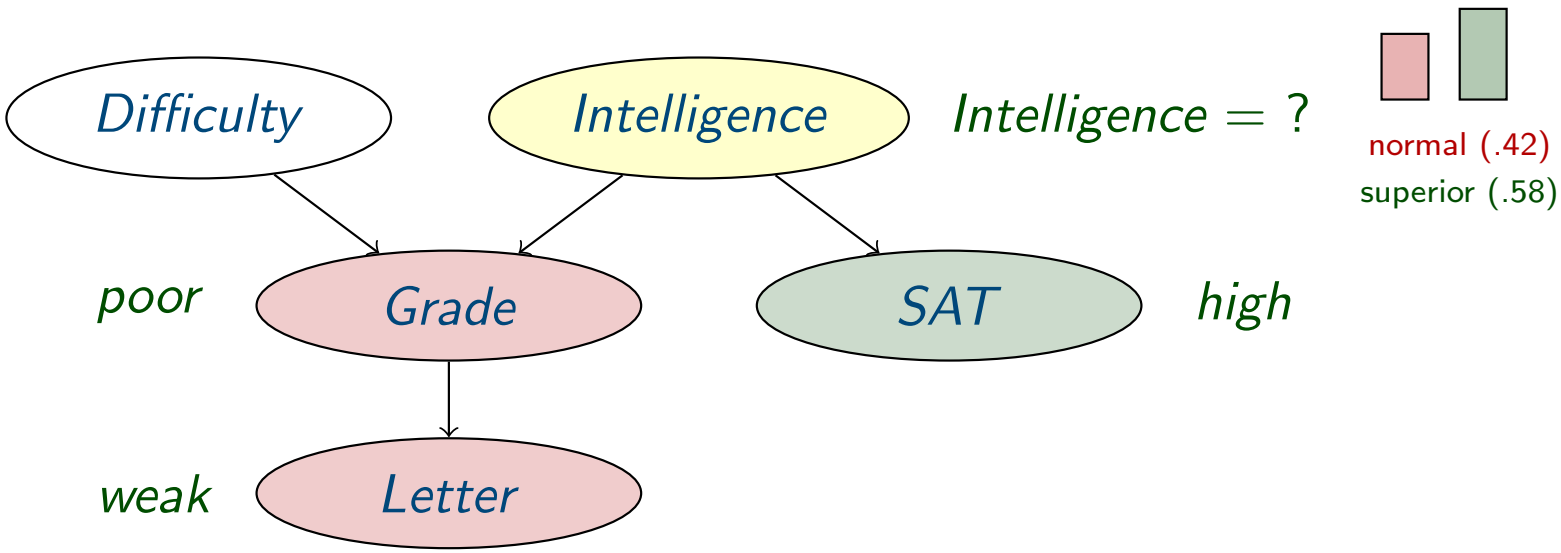
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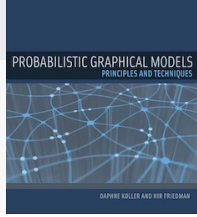




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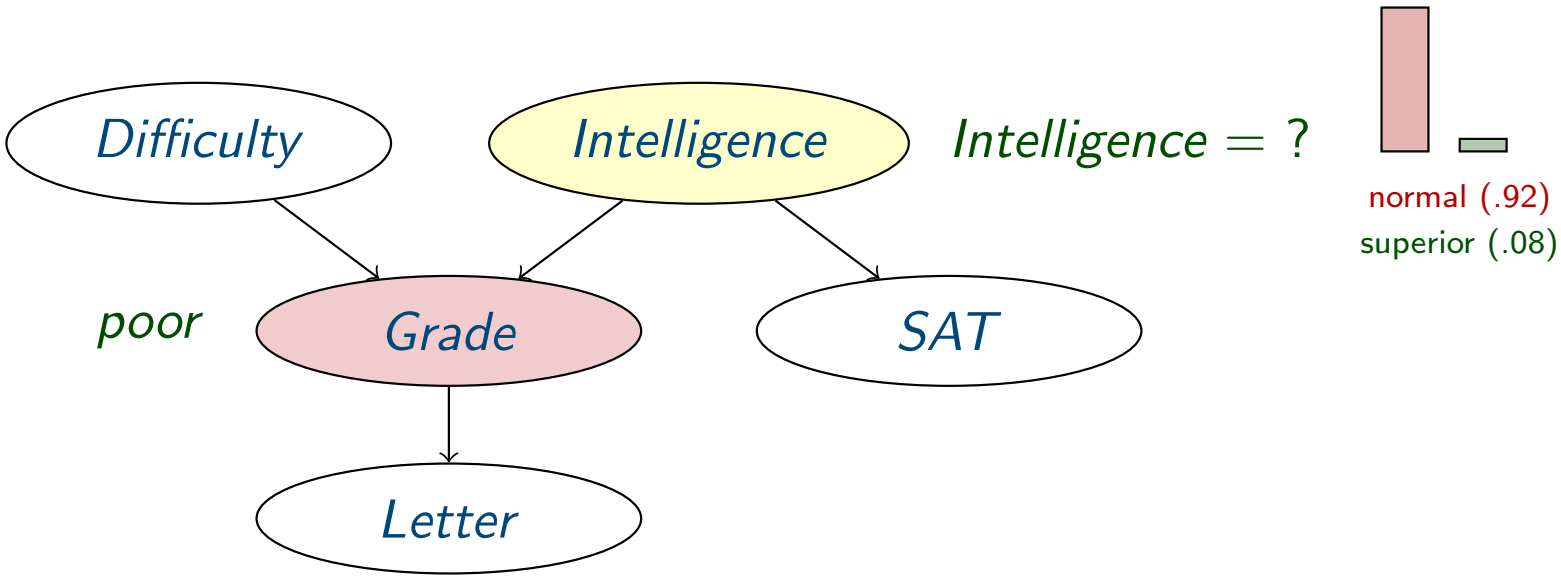
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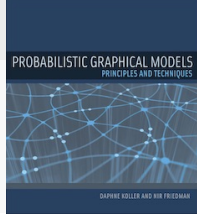




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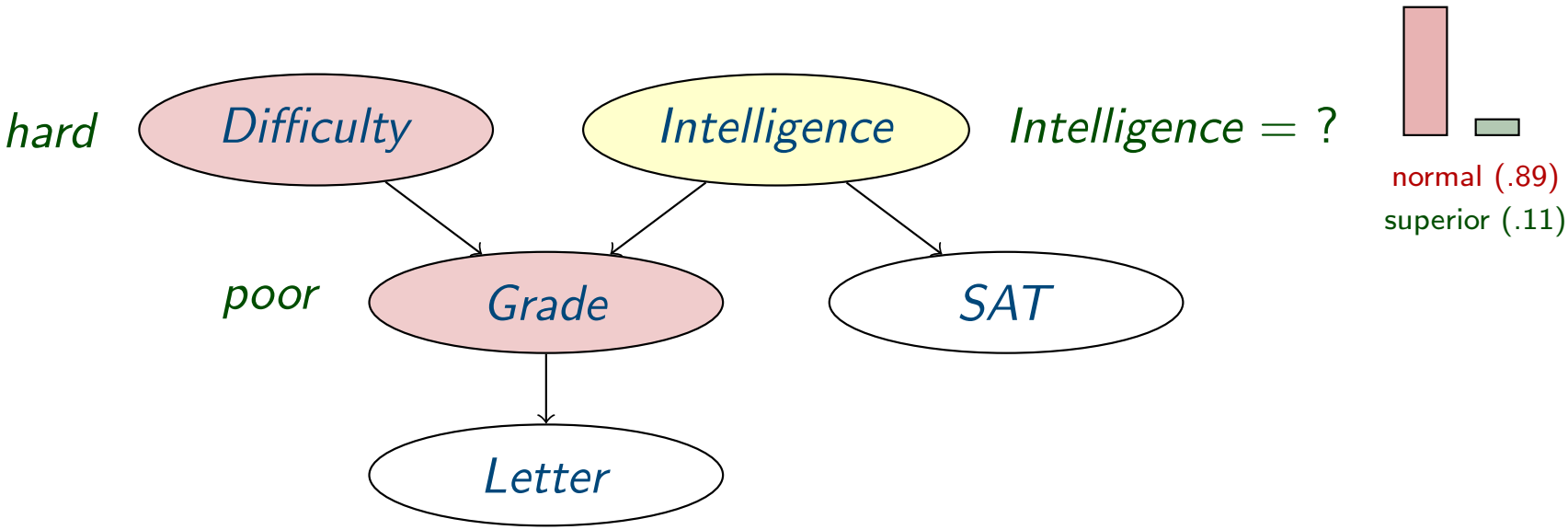
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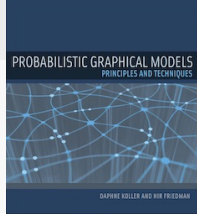




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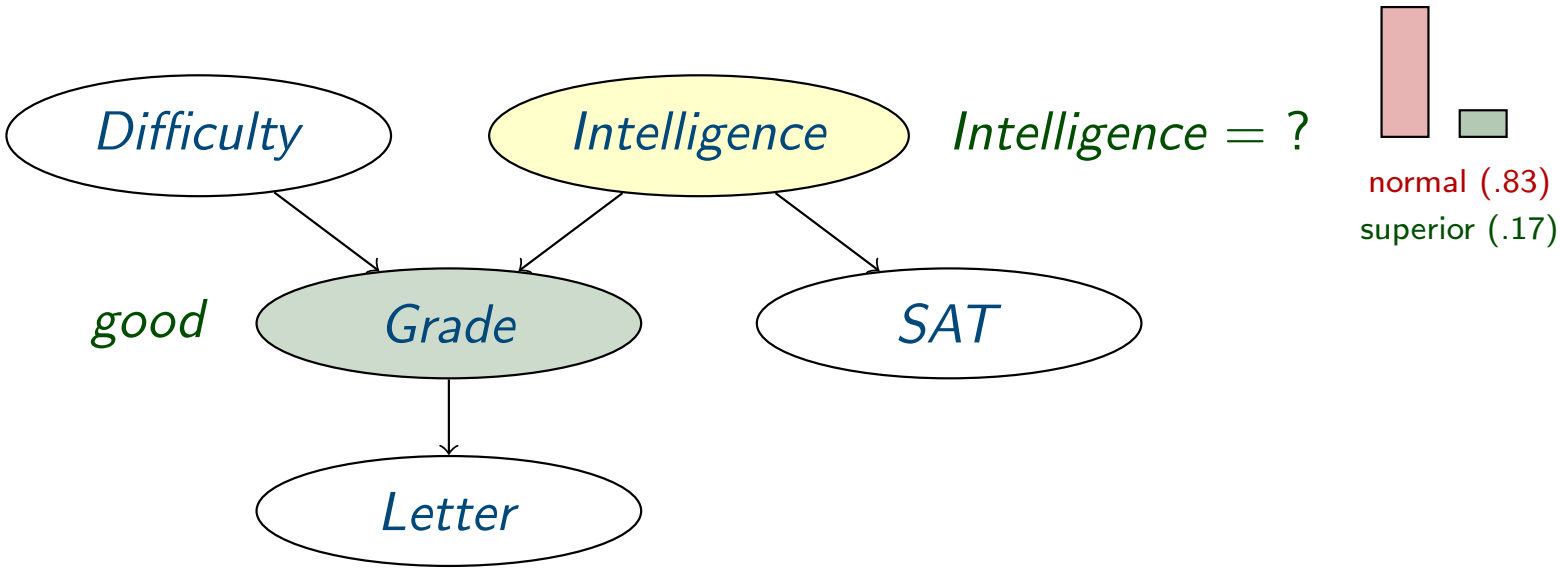
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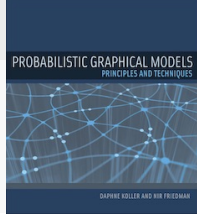




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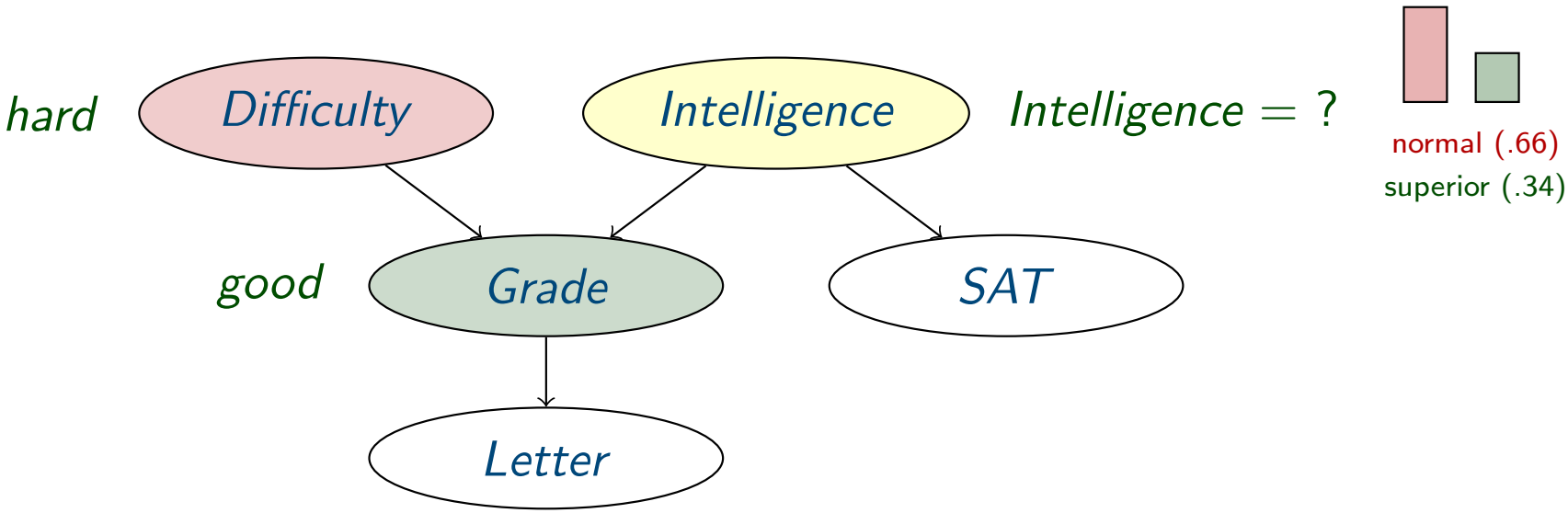
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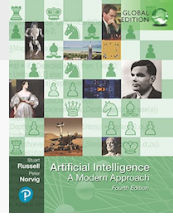




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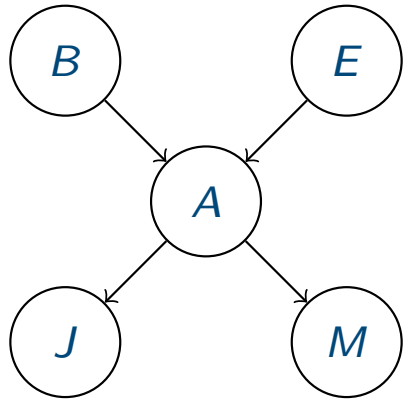
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Compactness

A CPT for Boolean X_i with k Boolean parents has 2^k rows for the combinations of parent values
Each row requires one number p for $X_i = \mathit{true}$
(the number for $X_i = \mathit{false}$ is just $1 - p$)

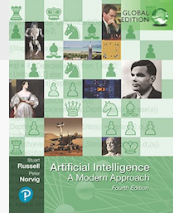


If each variable has no more than k parents, the complete network requires $O(n \cdot 2^k)$ numbers

i.e., grows linearly with n, vs. $O(2^n)$ for the full joint distribution

For burglary net, $1 + 1 + 4 + 2 + 2 = 10$ numbers (vs. $2^5 - 1 = 31$)

For student net, $1 + 1 + 8 + 2 + 3 = 15$ numbers (vs. $2^4 \times 3 - 1 = 47$)



Global semantics

Global semantics of a Bayesian network

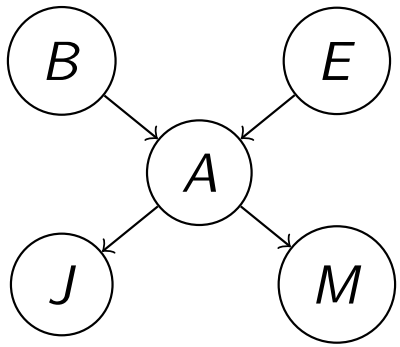
The local conditional probability distributions associated with all nodes jointly define the full joint probability distribution:

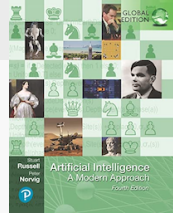
$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid \mathrm{pa}_i),$$

where pa_i denotes the values assigned to the parents of X_i in the complete assignment $(x_1, \ldots, x_n)$.

How to read the formula

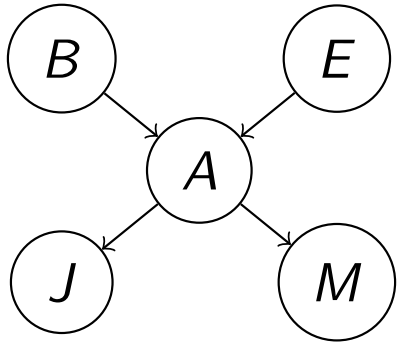
Compute the probability of one complete assignment: select one entry from the local CPT of every node and multiply all selected entries.





Global semantics: an atomic event

$$j \wedge m \wedge a \wedge \neg b \wedge \neg e.$$

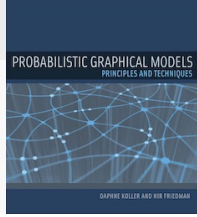


The alarm is sounding; John and Mary call; there is neither a burglary nor an earthquake.

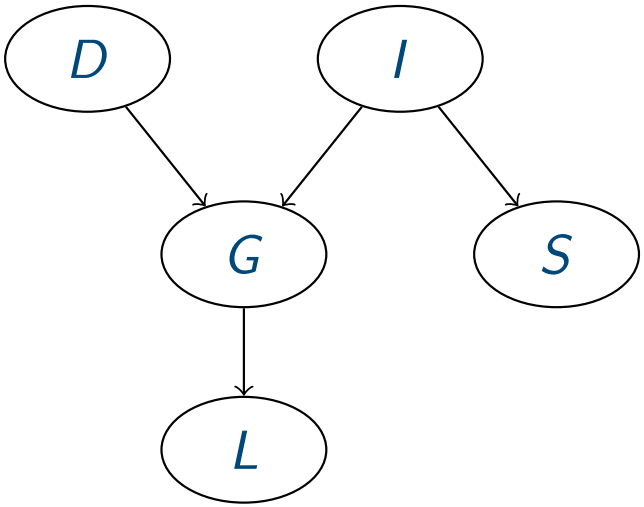
The global semantics tells us to select one local probability for each node:

$$\begin{aligned} P(j, m, a, \neg b, \neg e) &= P(j \mid a)\, P(m \mid a)\, P(a \mid \neg b, \neg e)\, P(\neg b)\, P(\neg e) \\ &= 0.9 \cdot 0.7 \cdot 0.001 \cdot 0.999 \cdot 0.998 \approx 0.00063. \end{aligned}$$

The Bayesian network reconstructs the probability of every atomic event from its local CPTs. We do not need to store the full joint distribution explicitly.

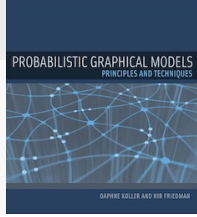


Basic independences in the Student network

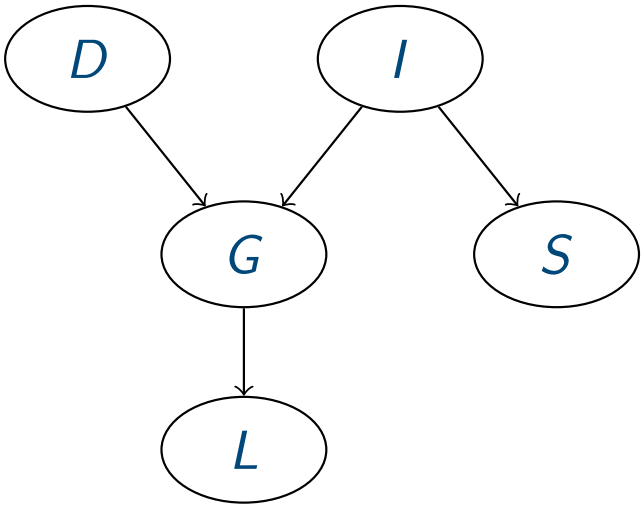


What independences?

- $\mathbf{P} \models (L \perp \ldots ?$
- $\mathbf{P} \models (S \perp \ldots ?$
- $\mathbf{P} \models (G \perp \ldots ?$
- $\mathbf{P} \models (I \perp \ldots ?$
- $\mathbf{P} \models (D \perp \ldots ?$

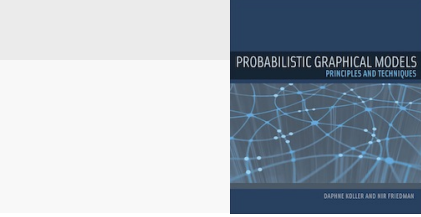


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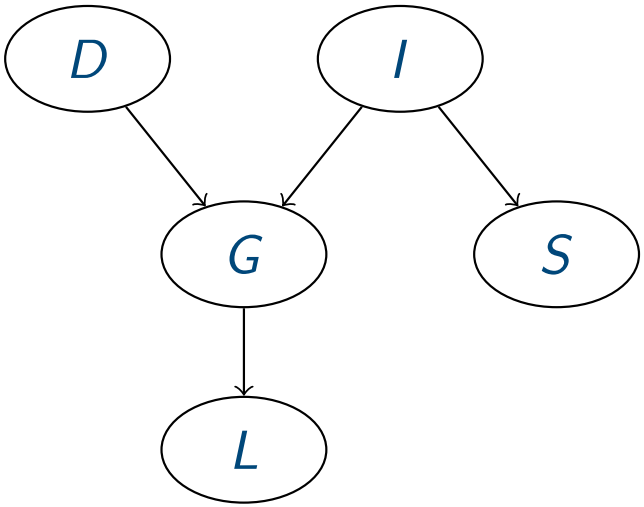


What independences?

- $\mathbf{P} \models (L \perp I, D, S | G)$
- $\mathbf{P} \models (S \perp G, D, L | I)$
- $\mathbf{P} \models (G \perp S | I)$
- $\mathbf{P} \models (I \perp D)$
- $\mathbf{P} \models (D \perp I, S)$
- ...



Flow of probabilistic influence

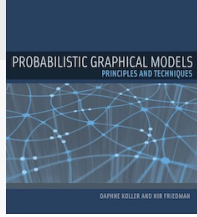


Or else, when could X influence Y?

- $X \rightarrow Y$ (direct cause)
- $X \leftarrow Y$ (direct effect)
- $X \rightarrow Z \rightarrow Y$ (causal trail)
- $X \leftarrow Z \leftarrow Y$ (evidential trail)
- $X \leftarrow Z \rightarrow Y$ (common cause)
- $X \rightarrow Z \leftarrow Y$ (common effect)

Intuition (active two-edge trail)

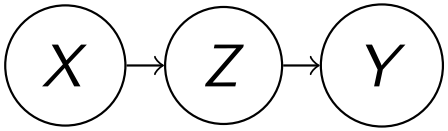
If influence can flow from X to Y via Z, the trail $X \rightleftharpoons Z \rightleftharpoons Y$ is active



Three basic trail patterns

A trail can be blocked or activated by observing intermediate variables.
The effect depends on the local structure of the trail.

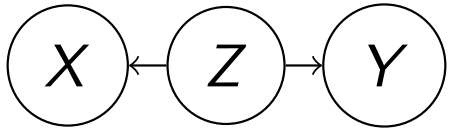
Chain



Z is a non-collider.

Z unobserved: active
Z observed: blocked

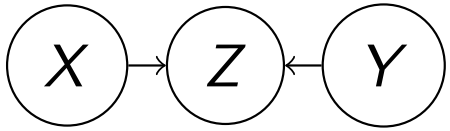
Common cause



Z is a non-collider.

Z unobserved: active
Z observed: blocked

Common effect

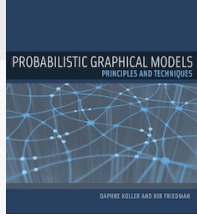


Z is a collider.

Z unobserved: blocked
Z observed: active

Intuition

- Observing an ordinary intermediate node blocks a trail.
- Observing a collider has the opposite effect: it opens a trail that was previously blocked.



Why can observing a node block or open a trail? I

Chain: observing the middle variable blocks the trail

$$Rain \rightarrow WetGrass \rightarrow SlipperyPath$$

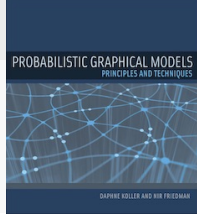
- *WetGrass* not observed: learning that it rained changes our belief that the path is slippery.
- *WetGrass* observed: knowing whether it rained provides no additional information about *SlipperyPath* ($Rain \perp SlipperyPath \mid WetGrass$.)



Wet grass unobserved: active trail



Wet grass observed: blocked trail

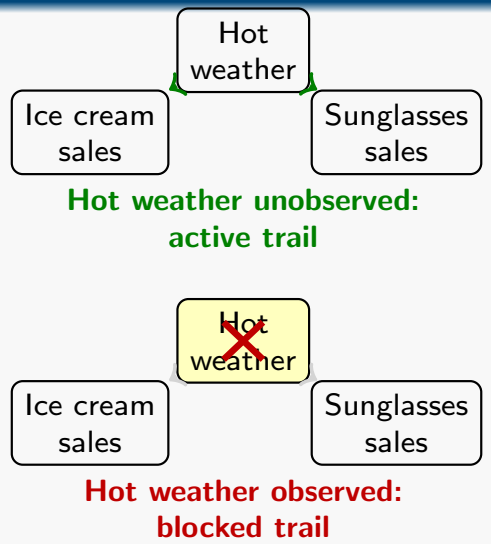


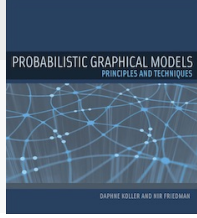
Why can observing a node block or open a trail? II

Common cause: observing the common cause blocks the trail

$IceCreamSales \leftarrow HotWeather \rightarrow SunglassesSales$

- *HotWeather* not observed: learning that ice-cream sales are high makes hot weather more plausible and therefore changes our belief that sunglasses sales are high.
- *HotWeather* observed: ice-cream sales provide no additional information about sunglasses sales ($IceCreamSales \perp SunglassesSales \mid HotWeather$.)



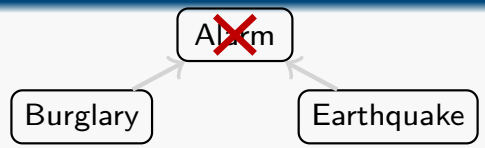


Why can observing a node block or open a trail? III

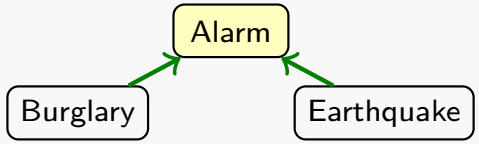
Collider: observing the common effect opens the trail

$$Burglary \rightarrow Alarm \leftarrow Earthquake$$

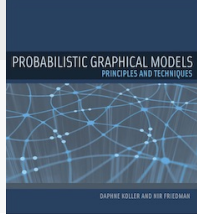
- Alarm not observed: burglary and an earthquake are independent ($Burglary \perp Earthquake$.)
- Alarm observed: evidence for one explanation changes the belief in the other. For example, if a burglary is confirmed, an earthquake becomes less necessary to explain the alarm



Alarm unobserved: blocked trail



Alarm observed: active trail

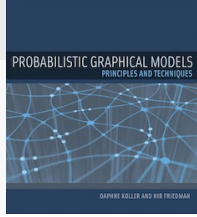


Active trails I

Trail

A trail is a path through a Bayesian network obtained by following connected nodes, regardless of the direction of the arrows.

Determine if 2 variables can be probabilistically dependent after observing some evidence: ask whether there is an active trail between them.



Active trails II

Active trail

Let Z be a set of observed variables. A trail $X_1 \rightleftharpoons \cdots \rightleftharpoons X_n$ is active given Z if every internal node X_i satisfies:

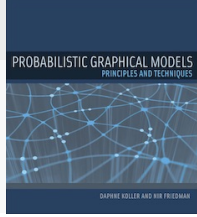
- if $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$, then X_i, or one of its descendants, $\in Z$;
- otherwise, $X_i \notin Z$.

How to read the two conditions

For every intermediate node on the trail:

- if it is a collider, then X_i, or one of its descendants, must be observed;
- every other intermediate node must **not** be observed.

- Observing a non-collider blocks a trail.
- Observing a collider (or one of its descendants) opens a trail.



D-separation I

D-separation

Two sets of nodes **X**, **Y** are d-separated given **Z** if there is no active trail between any $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ given **Z**

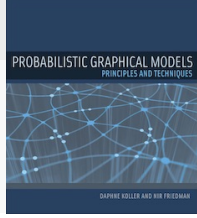
Why d-separation matters

For every probability distribution represented by the Bayesian network:

$$X \text{ and } Y \text{ d-separated given } Z \quad \Longrightarrow \quad X \perp Y \mid Z.$$

To determine if X and Y are **independent** given **Z**:

1. traverse the graph bottom-up marking all nodes in **Z** or having descendants in given **Z**
2. traverse the graph from X to Y, stopping if we get to a blocked node
3. if we can't reach Y, then X and Y are independent



D-separation II

Blocked node

Given a set of observed variables **Z**, an intermediate node on a trail is blocked if either:

- it is the middle node of an unmarked v-structure (i.e., a collider for which neither the node nor any of its descendants is observed); or
- it is a non-collider that belongs to **Z**.



A guided d-separation example I

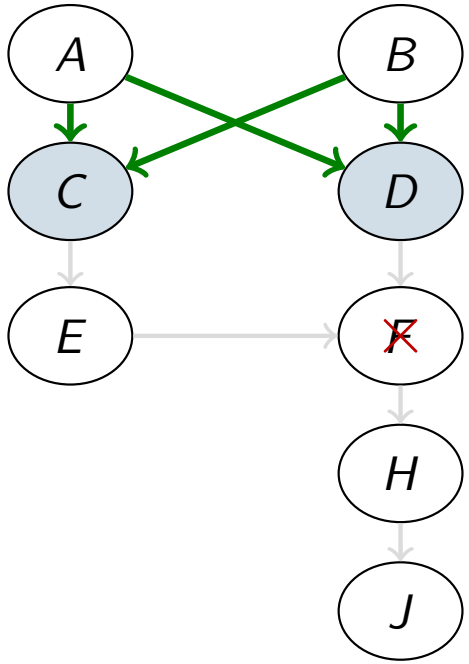
For each set of observed variables $\mathbf{Z}$, determine whether C and D are d-separated given $\mathbf{Z}$.

The relevant trails are:

$$T_1 : C \leftarrow A \rightarrow D$$
$$T_2 : C \leftarrow B \rightarrow D$$
$$T_3 : C \rightarrow E \rightarrow F \leftarrow D$$

T_1 and T_2 contain common causes.

T_3 contains the collider F.



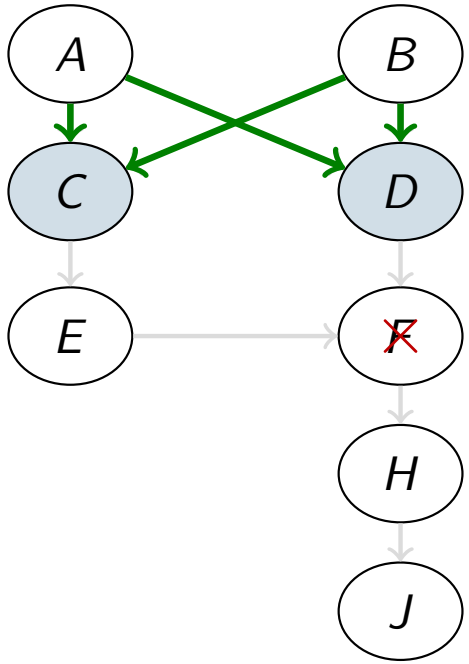
Green: active trail **×: blocking node**



A guided d-separation example II

Step 1 – No observations

$$C \perp D \, ?$$



C and D are not independent.

The trails through A and B are active.

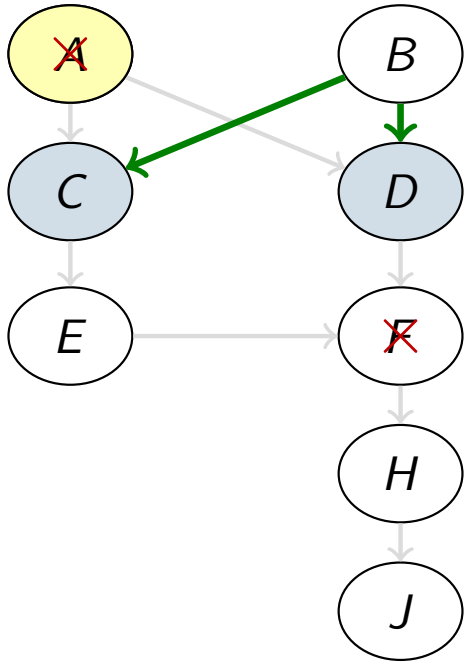
The collider trail through F remains blocked because neither F nor one of its descendants is observed.



A guided d-separation example III

Step 2 – Observe A

$$C \perp D \mid A \, ?$$



C and D are not independent.

Observing the non-collider A blocks T_1.

However, T_2, through the unobserved common cause B, is still active.



A guided d-separation example IV

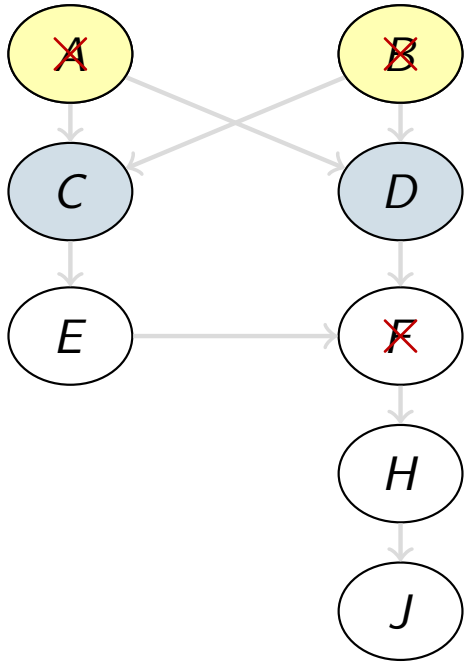
Step 3 – Observe A and B

$$C \perp D \mid A, B \ ?$$

C and D are independent.

Observing A and B blocks both common-cause trails.

The remaining trail is still blocked at the unobserved collider F.

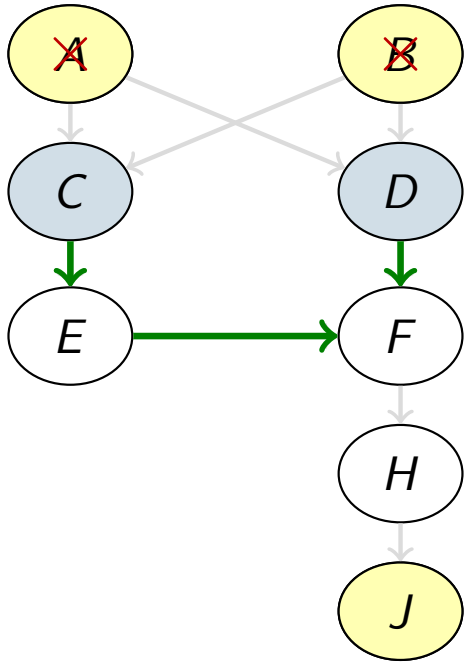




A guided d-separation example V

Step 4 – Observe A, B, and J

$$C \perp D \mid A, B, J \, ?$$



C and D are not independent.

J is a descendant of the collider F.

Observing J opens the collider trail:

$$C \rightarrow E \rightarrow F \leftarrow D.$$



A guided d-separation example VI

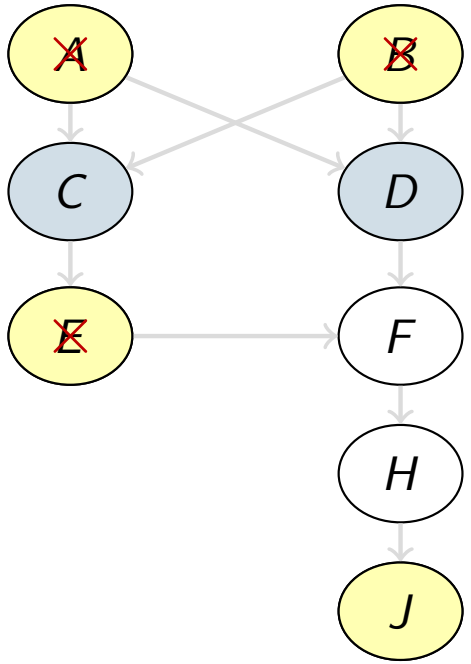
Step 5 – Observe A, B, E, and J

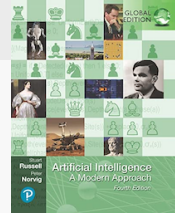
$$C \perp D \mid A, B, E, J \ ?$$

C and D are independent.

Observing J opens the collider at F, but observing E, a non-collider on that trail, blocks it again.

All trails between C and D are now blocked.

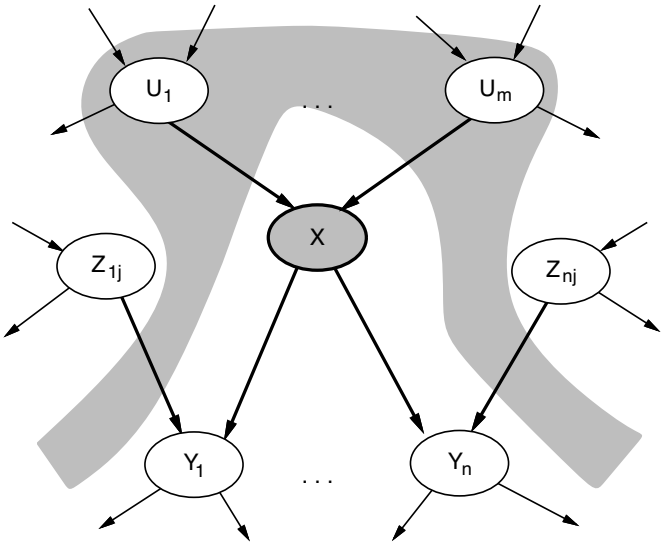




Local semantics I

Local semantics (local Markov property)

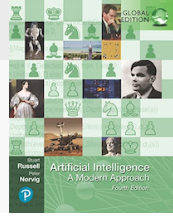
For every node X, once its parents are known, X is conditionally independent of all its non-descendants that are not parents: $X \perp (\text{NonDescendants}(X) \setminus \text{Parents}(X)) \mid \text{Parents}(X)$.



How to interpret local semantics

To assess how likely a node is, it is sufficient to know the values of its direct parents. Once they are known, learning about any other non-descendant provides no additional information.

For example: *Burglary* $\rightarrow$ *Alarm* $\rightarrow$ *JohnCalls*. Once we know whether the alarm is sounding, learning whether a burglary occurred does not change our belief about *JohnCalls*: *JohnCalls* $\perp$ *Burglary* $\mid$ *Alarm*.

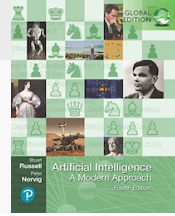


Local semantics II

Connection with global semantics

For a Bayesian network, the local Markov property is equivalent to the global factorization of the joint probability distribution.

Theorem: Local semantics $\Leftrightarrow$ global semantics



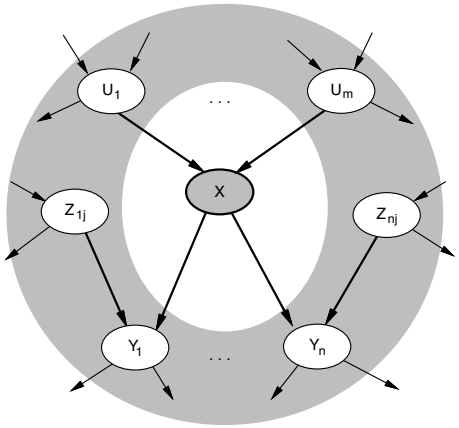
Markov blanket

Markov blanket

The Markov blanket of a node X consists of:

- the parents of X;
- the children of X;
- the other parents of the children of X.

Given its Markov blanket, X is conditionally independent of every other node in the network: $X \perp \left(\mathbf{V} \setminus (\{X\} \cup MB(X))\right) \mid MB(X)$.



$MB(Alarm) = \{Burglary, Earthquake, JohnCalls, MaryCalls\}$. It contains the parents and children of $Alarm$. In this network, the children have no other parents.

MB is the smallest local set of variables that contains all the information about a node available in the network.

From questions: What is a directed acyclic graph? I

Bayesian network: syntax

A Bayesian network for a set of random variables $X_1, \ldots, X_n$ consists of:

- one node for each variable;
- a directed acyclic graph (DAG), i.e., a graph with directed edges and no directed cycles;
- one local conditional probability distribution for each node:

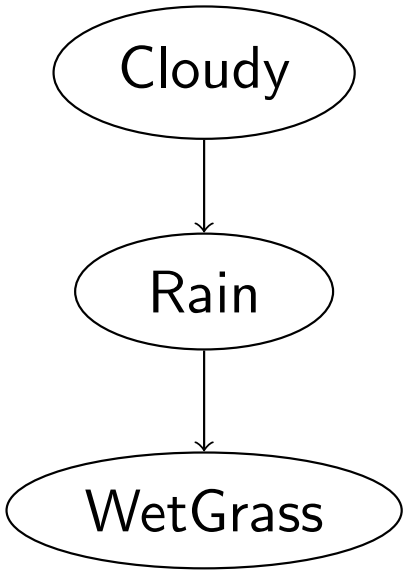
$$P(X_i \mid \mathrm{Parents}(X_i)).$$

From questions: What is a directed acyclic graph? II

Directed acyclic graph (DAG)

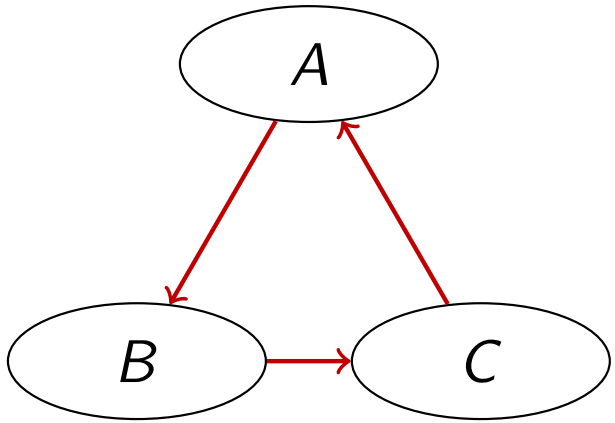
A graph whose edges have a direction and in which it is impossible to start from a node and return to it by following the direction of the arrows.

Valid: a DAG



Following the arrows, we can never return to a node already visited.

Not valid: directed cycle



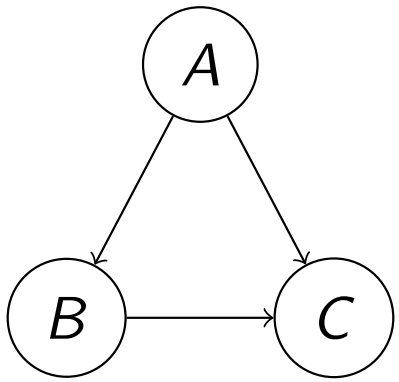
Following the arrows gives $A \to B \to C \to A$: this is a directed cycle.

From questions: What is a directed acyclic graph? III

Why is acyclicity required?

It lets us order the variables so that the probability of each node is specified only from variables that come earlier in the ordering.

Still valid: no directed cycle



The underlying graph has a triangle, but there is no path that follows the arrows and returns to its starting node.

Questions?

